Project Proposal

Only for course Teacher						
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Allocate mark & Percentage		25%	50%	75%	100%	15
Clarity	3					
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Semester: Fall……2026……

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Batch: 42 **Section: A1** **Course Code: SE-331**

Course Name: Software Engineering Design Capstone Project

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Submission Date: …02…. /…10…/…2026….

Project Title

FabricIQ AI – Intelligent Fabric Inspection, Quality Analysis & Design Recommendation System

1. Introduction

Fabric inspection, quality assessment, and design selection are important processes in the textile and garment industry. These activities help manufacturers maintain product quality, reduce material waste, and select suitable fabrics and designs for different products. However, many of these tasks still depend on manual inspection, personal experience, and separate tools.

FabricIQ AI is a proposed AI-powered web-based platform designed to automate and assist different stages of fabric analysis. The system will use computer vision, image classification, optical character recognition (OCR), and AI-based recommendation techniques to analyze fabric images, detect visible defects, estimate fabric characteristics, recommend suitable designs, and generate inspection reports.

The system will support both live-camera inspection and image uploading, allowing users to analyze fabric through a centralized platform. Its primary goal is to reduce repetitive manual work and provide faster, more consistent, and data-supported results while keeping final decisions under human supervision.

1.1 Background Overview

The textile and garment industry involves several activities, including fabric inspection, material identification, quality assessment, design planning, and product development. Traditionally, quality inspectors examine fabric visually to identify defects such as stains, holes, stitching problems, and surface irregularities. Designers and production managers also rely on their experience, reference catalogs, and market information to select suitable fabric types, colors, and patterns.

Although some modern industrial systems use AI-based computer vision for automated defect detection, many workflows still involve manual inspection and separate systems for recording results, analyzing fabric properties, and selecting designs. These disconnected processes may increase inspection time and make it difficult to maintain consistent records.

Artificial Intelligence can assist these processes by identifying visual patterns, detecting defects, classifying supported fabric appearances, and comparing fabric characteristics with existing design references. Computer vision models can analyze images, while OCR can extract readable information from garment care labels. Recommendation techniques can use analyzed fabric attributes to suggest relevant design references.

FabricIQ AI aims to integrate these AI capabilities into a single platform. The system will focus on visual fabric analysis and AI-assisted recommendations rather than claiming to replace laboratory testing or professional textile expertise.

1.2 Problem Statement

Fabric inspection and quality assessment in the garment industry often require considerable manual effort. Inspectors must visually examine fabric, locate defects, classify their types, estimate the visible condition, and document the results. These tasks can be repetitive and may produce inconsistent outcomes depending on individual experience, inspection conditions, and workload.

In addition to inspection, selecting suitable designs and prints for particular fabric types, colors, and patterns can require time-consuming manual comparison. Care instructions may also need to be interpreted separately from product labels, while market and seasonal decisions may depend on information collected from different sources.

Existing manual practices and separate digital tools do not necessarily provide a unified workflow for these activities. This creates a need for a software-based solution that can centralize fabric inspection, organize results, and assist users with visual analysis and recommendations.

AI is particularly relevant to image-based tasks, such as identifying visible defects, estimating supported fabric categories, analyzing color and pattern, and retrieving visually similar design references. These tasks involve visual characteristics that are difficult to handle efficiently using fixed rules alone.

The proposed system will focus on a limited set of visible fabric defects and supported fabric categories. It will not claim to identify exact fiber composition, guarantee fabric durability, determine an exact maximum number of washes from an image, or predict market demand without reliable supporting data.

1.3 Objectives

General Objective

To develop an AI-powered web-based platform that automates and assists fabric inspection, visual quality analysis, fabric appearance identification, design recommendation, care-label interpretation, and inspection reporting.

AI Objective

The specific AI objectives of the proposed system are:

- To develop a computer vision model that detects and localizes selected visible fabric defects from live-camera frames or uploaded images.
- To classify supported fabric appearance categories using image-based machine learning techniques.
- To analyze dominant fabric colors, textures, and visible patterns.
- To generate an AI-assisted visual inspection score based on detected defects and defined image-analysis criteria.
- To recommend suitable design and print references using fabric characteristics and a curated design library.
- To use OCR to extract readable care-label information and assist in interpreting washing and care instructions.
- To automate the generation of structured inspection summaries and support data-driven decisions.

1.4 Scope

In-Scope Features

The proposed system will include the following features:

- User registration, login, and role-based access.
- Live-camera fabric inspection using a webcam or compatible camera.
- Fabric image upload for AI-based analysis.
- Detection and localization of selected visible fabric defects, such as stains, holes, and supported stitching or surface defects.
- Image-based estimation of supported fabric categories, such as cotton, denim, linen, and polyester, subject to dataset availability.
- Fabric color, texture, and visible pattern analysis.
- AI-assisted visual quality scoring based on documented criteria.
- AI-based design and print recommendations from a curated design-reference library.
- Similar design retrieval based on visual characteristics.
- Care-label text extraction using OCR and interpretation of supported instructions.
- Basic season- and occasion-based design recommendations using available reference data.
- Automatic inspection report generation.
- Inspection history, defect statistics, and dashboard analytics.
- Human review and correction of AI-generated results.

Out-of-Scope Features

The initial version of the system will not include:

- Laboratory-based identification of exact fiber composition.
- Physical testing of fabric strength, chemical properties, or colorfastness.
- Guaranteed prediction of the maximum number of washing cycles.
- Certified industrial quality approval or automatic rejection of production batches.
- Automated control of industrial sewing, weaving, or printing machines.
- Guaranteed live market-demand forecasting without verified and regularly updated market data.
- Detection of every possible fabric defect or recognition of all fabric types.
- Unrestricted downloading or scraping of copyrighted designs from websites such as Pinterest.

AI Boundaries

The AI model will perform visual defect detection, supported fabric appearance classification, color and pattern analysis, and design-reference recommendations. OCR will be used for readable care-label information.

The AI will provide estimates and recommendations based on the available dataset and system information. It will not replace laboratory testing, verified manufacturer specifications, or the final judgment of a qualified inspector. Predictions with low confidence will be identified for human review.

The prototype will initially be evaluated under controlled camera distance, lighting, and sample conditions to keep the project achievable.

1.5 Stakeholders

Stakeholder	Role and Interest
Quality Inspectors	Use the system to inspect fabric, review AI-detected defects, confirm results, and maintain inspection records.
Production Managers	Monitor defect statistics, review inspection reports, and identify recurring quality issues.
Textile and Fashion Designers	Use fabric analysis and design recommendations to support design planning.
Factory Administrators	Manage user accounts, system settings, defect categories, and design-reference data.
Garment Factories	Potential organizational users interested in reducing repetitive inspection work and improving record management.
Project Team	Design, develop, test, and maintain the proposed system and evaluate its AI performance.
Project Supervisor	Review the project's feasibility, progress, technical implementation, and academic outcomes.

1.6 Proposed Solution

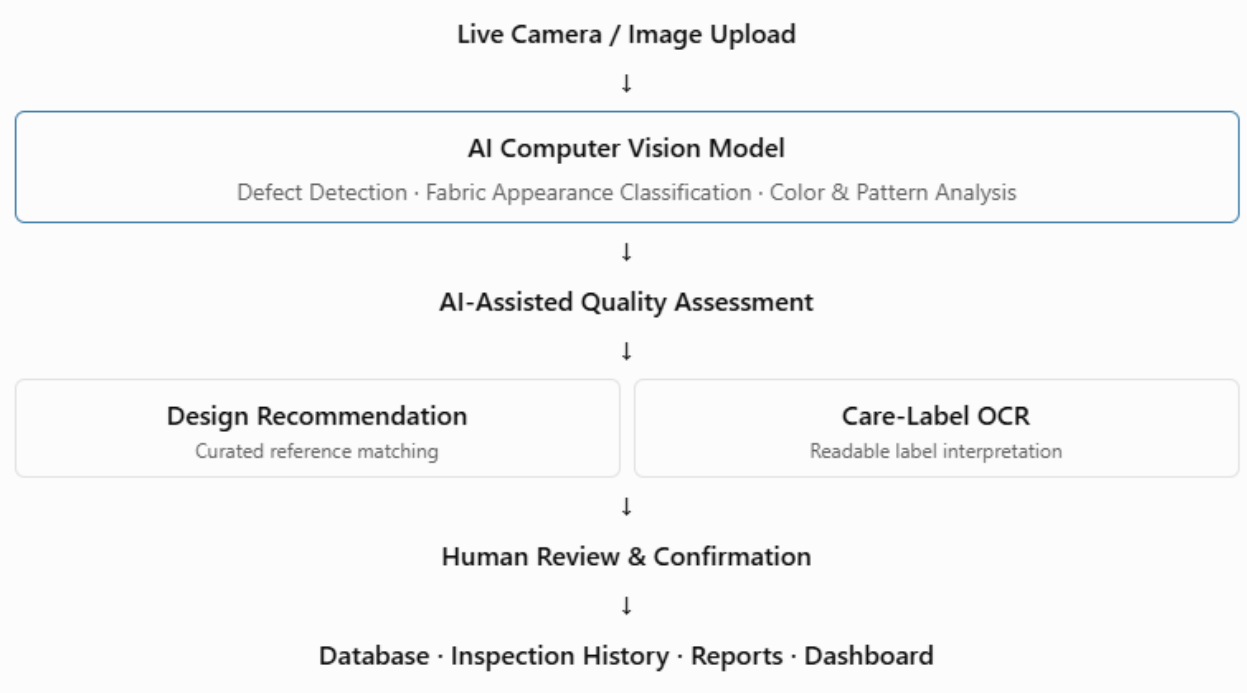
FabricIQ AI will provide a centralized web-based platform that combines AI-based fabric inspection and related decision-support features.

A user will log in to the system and either place fabric in front of a live camera or upload an image. The backend will process the image and send it to the appropriate AI model. The computer vision model will identify selected visible defects and return their locations, predicted classes, and confidence scores.

Additional image-analysis modules will estimate supported fabric categories, dominant colors, textures, and visible patterns. The system will use these characteristics to recommend suitable design references from a curated library. If the user provides a care-label image, OCR will extract readable text to support care-instruction interpretation.

The user will be able to review, confirm, or correct the results before saving them. The system will store inspection information in a database and provide inspection history, analytics, and automatically generated reports through a dashboard.

System Workflow



The AI components will perform the central image-analysis and recommendation tasks, while the web application will manage user access, records, reviews, and reporting.

1.7 AI Features

Existing AI Features

AI techniques such as computer vision, object detection, image classification, OCR, and image similarity search are already used in various image-analysis and industrial inspection applications. Computer vision can identify visual patterns and localize defects, while OCR can extract text from suitable images. Image embeddings can also be used to find visually similar references.

However, these techniques have limitations. Model performance depends on dataset quality, image resolution, lighting, camera position, and the range of defect or fabric classes represented in training data. Visual appearance alone cannot reliably determine all physical properties of a fabric. Similarly, recommendation systems require suitable reference data to provide relevant results.

FabricIQ AI will adapt selected AI techniques to a controlled fabric-inspection prototype and evaluate their performance using a dedicated test dataset.

Proposed AI Features

AI Feature	Input	Output	Manual Work Supported or Reduced
Fabric Defect Detection	Live-camera frame or fabric image	Defect class, location, and confidence score	Repetitive visual scanning and defect marking
Fabric Type Estimation	Fabric image	Predicted supported fabric appearance category	Initial manual categorization
Color and Pattern Analysis	Fabric image	Dominant colors and visible pattern labels	Manual visual description and tagging
Visual Quality Assessment	Defect results and image indicators	Preliminary visual inspection score	Repetitive preliminary assessment
Design Recommendation	Fabric attributes and reference library	Suitable design and print references	Manual browsing and comparison
Similar Design Search	Fabric image or selected attributes	Visually similar design references	Searching through reference collections
Care-Label OCR	Care-label image	Extracted readable text and supported interpretation	Manual transcription and explanation
Automated Reporting	Reviewed inspection data	Structured inspection report	Manual report preparation

AI Techniques

The project will use the following AI techniques:

- Object Detection: A YOLO-family model or an equivalent object-detection architecture will be trained or fine-tuned to detect selected visible fabric defects and localize them with bounding boxes.
- Image Classification: A suitable classification model will estimate supported fabric appearance categories based on visual characteristics.
- Image Processing: Image-processing techniques will assist in analyzing dominant colors and visible patterns.
- Optical Character Recognition (OCR): An OCR library will extract readable text from care labels.
- Image Similarity Search: Image embeddings and similarity comparison will retrieve relevant design references from a curated collection.
- Recommendation Logic: AI-derived fabric attributes and reference metadata will be used to recommend suitable design options.
- Rule-Based Scoring: A transparent scoring method will combine defined visual indicators into a preliminary inspection score.
- Optional Generative AI: A generative model may be explored to create design concepts as an advanced feature, depending on available time and resources.

Why AI Is Essential

AI is essential to the core functionality of FabricIQ AI because the system is intended to automate image-based fabric inspection and analysis. Without AI, it would not be able to automatically detect and localize visible defects, estimate supported fabric categories, analyze visual patterns, or retrieve relevant design references based on fabric attributes.

If all AI components were removed, the platform would mainly function as a manual record-management system. The primary automated inspection and recommendation capabilities would no longer be available. Therefore, AI is integrated into the central workflow rather than being used only as an additional chatbot or optional feature.

2. System Requirements

This section describes the main functional and non-functional requirements of FabricIQ AI. The requirements are defined to make the system's expected behavior measurable and testable.

2.1 Functional Requirements

ID	Requirement
FR-01	The system shall allow users to register and log in using authorized credentials.
FR-02	The system shall provide role-based access for inspectors, managers, and administrators.
FR-03	The system shall allow users to capture fabric images using a supported live camera or upload images for inspection.
FR-04	The system shall process submitted images and return AI analysis results for supported fabric and defect categories.
FR-05	The system shall display detected defects with their class, bounding box, and confidence score where available.
FR-06	The system shall display estimated fabric category, dominant colors, and supported visible patterns.
FR-07	The system shall generate a preliminary visual quality score using documented criteria.
FR-08	The system shall allow inspectors to review, confirm, or correct AI-generated findings before saving inspection results.
FR-09	The system shall recommend design references based on analyzed fabric characteristics and the available reference library.
FR-10	The system shall accept care-label images and extract readable text using OCR.
FR-11	The system shall store inspection records, including date, user, image reference, AI results, and reviewed outcome.

FR-12	The system shall allow authorized users to view previous inspection records and basic defect statistics.
FR-13	The system shall generate a structured inspection report from saved inspection data.
FR-14	The system shall identify low-confidence or unsupported predictions and indicate when human review is required.
FR-15	The system shall allow administrators to manage users, supported defect categories, fabric categories, and design references.

2.2 Non-Functional Requirements

ID	Requirement
NFR-01	The system shall provide a user-friendly interface accessible through common desktop and laptop browsers.
NFR-02	The prototype shall target a processing time of up to 5 seconds for a single still image on the team's test hardware, subject to model complexity and hardware capability.
NFR-03	The system shall validate image formats and file sizes before processing.
NFR-04	The system shall store passwords securely using password hashing rather than plain text.
NFR-05	The system shall restrict inspection records and administrative functions according to user roles.
NFR-06	The system shall communicate that AI outputs are estimates and require human review where appropriate.
NFR-07	The AI model shall be evaluated using suitable metrics, such as precision, recall, and mean Average Precision (mAP) for object detection.
NFR-08	The system shall display understandable error messages when camera access, image upload, or AI processing fails.
NFR-09	The system shall maintain inspection history and allow authorized users to retrieve saved records.
NFR-10	The system shall be designed in modules so that AI models and recommendation features can be updated independently.

3. Tools and Technologies

The following technologies are proposed for developing FabricIQ AI. Final selections may be adjusted based on feasibility tests, available hardware, and the project team's experience.

Category	Proposed Technology	Purpose
Programming Language	Python	AI model development, image processing, and backend logic
Frontend	React.js, JavaScript	User interface, live camera, dashboard, and result visualization
Backend	FastAPI	API development, authentication, image processing, and AI integration
Computer Vision	OpenCV	Image preprocessing and camera-frame handling
AI Model	YOLO-family model, PyTorch	Defect detection and localization
Image Classification	PyTorch / suitable pretrained model	Supported fabric appearance classification
OCR	PaddleOCR or Tesseract	Care-label text extraction
Image Similarity	Image embeddings and similarity search	Matching fabric attributes with design references
Database	MySQL	User data, inspection records, and AI results
Authentication	JWT and secure password hashing	User authentication and role-based access
Frontend Styling	CSS / Tailwind CSS	Responsive interface and visual presentation
Version Control	Git and GitHub	Source-code management and team collaboration
API Testing	Postman	Testing backend endpoints
Deployment	Docker and suitable hosting	Application packaging and deployment

4. Project Timeline and Work Plan

The project is planned over a period of 12 weeks. The schedule is a proposed development plan and may be adjusted according to the official capstone calendar and supervisor feedback.

4.1 Milestone-Based Timeline

Milestone	Duration	Activities	Expected Deliverable
Milestone 1: Requirement Analysis and Planning	Weeks 1–2	Define project scope, identify user needs, review related work, prepare use cases, and design the initial architecture.	Requirement specification and initial system design
Milestone 2: Dataset Preparation	Weeks 3–4	Collect permitted fabric images, define defect classes, annotate images, and prepare training, validation, and test datasets.	Labeled fabric dataset
Milestone 3: AI Model Development	Weeks 5–6	Train or fine-tune the defect detection model, test fabric classification feasibility, and evaluate baseline performance.	Initial AI models and evaluation results
Milestone 4: Backend and Database Development	Weeks 6–8	Develop APIs, authentication, database schema, image handling, and AI inference endpoints.	Functional backend and database
Milestone 5: Frontend and Camera Integration	Weeks 8–9	Develop the React interface, image upload, webcam workflow, result display, and dashboard.	Integrated frontend and AI inspection interface
Milestone 6: Recommendation and Reporting	Weeks 10–11	Implement design-reference matching, care-label OCR prototype, inspection history, analytics, and report generation.	Recommendation and reporting modules
Milestone 7: Testing and Finalization	Week 12	Perform functional and AI testing, fix issues, document limitations, deploy the prototype, and prepare the final demonstration.	Tested prototype and final project documentation

4.2 Team Task Distribution

The project work will be divided among three team members to ensure balanced workload, efficient development, and proper coordination.

Team Member	Assigned Role	Responsibilities
Md.Sajjad Hossain Nahid	Frontend Developer & UI/UX Designer	- Design and develop the user interface of the web application. - Build the dashboard, image upload, and live camera interface. - Display AI detection results, confidence scores, and inspection reports. - Ensure the application is user-friendly and responsive.
MD. Yasin	AI/ML Developer	- Collect, organize, and preprocess fabric images for the dataset. - Train and test AI models for fabric defect detection and fabric classification. - Implement color, texture, and pattern analysis where feasible. - Integrate the trained AI model with the backend and evaluate its performance.
Sumaiya akter Sumi	Backend Developer & Database Manager	- Develop backend APIs and connect the frontend with AI models. - Design and manage the database for user information, inspection results, and history. - Implement inspection report generation and data management. - Handle application deployment, backend testing, and system integration.

Shared Responsibilities:

- All members will participate in requirement analysis, project planning, and defining system features.
- All members will contribute to testing, debugging, documentation, and final presentation preparation.
- Team members will review each other's work and coordinate regularly to ensure successful integration.
- Each member will maintain regular GitHub commits and provide progress updates according to the project timeline.

5. Optional Sections

5.1 Existing System

Fabric inspection is commonly performed through manual visual examination, where inspectors identify defects and record their findings. In some workflows, digital spreadsheets or separate software tools are used to manage inspection data. Design selection and care-label interpretation may also be handled through independent resources.

Computer-vision-based approaches can support automated detection of visible defects, while OCR and image-retrieval techniques can assist with other image-related tasks. However, these capabilities may be offered separately and may require specialized data, equipment, or configuration.

FabricIQ AI proposes to bring selected inspection, fabric-image analysis, recommendation, and reporting capabilities together in one web-based prototype. Its purpose is to assist users with repetitive visual tasks and organize the resulting information rather than fully replace experienced textile professionals.

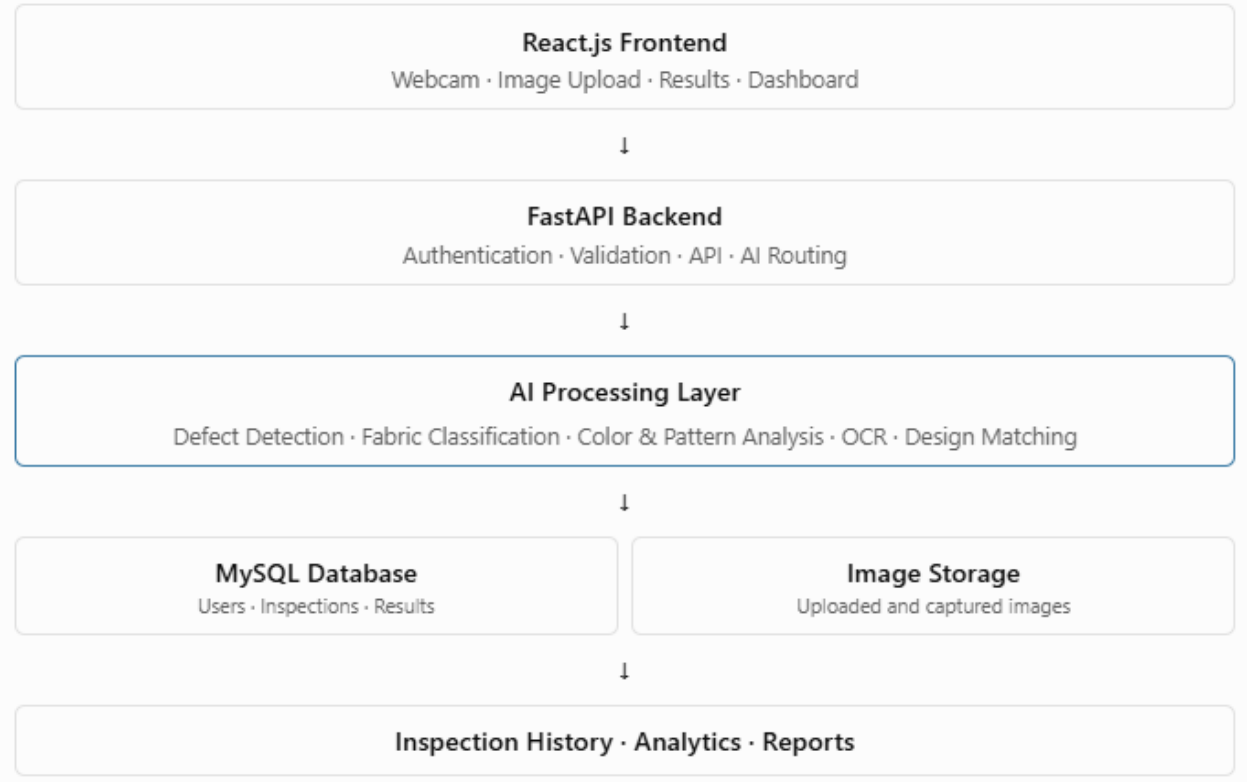
5.2 System Architecture

FabricIQ AI will use a modular client-server architecture. The React frontend will provide the user interface and capture images through a compatible webcam or upload function. The FastAPI backend will validate requests, manage authentication, and route images to the appropriate AI models.

The AI modules will perform defect detection, fabric appearance classification, color and pattern analysis, and design-reference matching. OCR will process care-label images when provided. The backend will organize the AI outputs and allow users to review them before saving.

MySQL will store user details, inspection metadata, AI predictions, review decisions, and report information. Image files will be stored using an appropriate file-storage solution, with database records retaining their references.

Architecture Overview



5.3 Ethical, Legal, and Social Considerations

The project will consider the following ethical, legal, and social aspects:

- **Data Privacy**: User accounts and inspection records will be protected through authentication and role-based access.
- **Dataset Permissions:** Images will be collected by the team or obtained from sources that permit the intended academic use. Dataset origins and permissions will be documented.
- **Copyright Compliance:** Design references will come from team-created, licensed, public-domain, or otherwise permitted sources. The system will not depend on unauthorized scraping or redistribution of copyrighted images.
- **AI Bias and Reliability:** Model performance may vary across fabric colors, patterns, textures, lighting conditions, and defect appearances. Testing will be used to identify limitations.
- **Transparency:** The system will show confidence information where available and distinguish estimated visual analysis from verified physical measurements.
- **Human Oversight**: Inspectors will be able to review and correct AI results. Final quality and production decisions will remain with responsible human users.
- **Responsible Recommendations:** Care and market-related recommendations will be based on available supporting information and will not be presented as guaranteed outcomes.

5.4 Future Work

The following features may be considered for future development, depending on the performance of the prototype and availability of resources:

- Expanding the dataset to include more fabric categories and defect types.
- Improving performance under different camera positions and lighting conditions.
- Integrating verified manufacturer data to improve fabric identification and care guidance.
- Incorporating reliable and permitted market or trend datasets for more evidence-based seasonal recommendations.
- Developing AI-generated design concepts and fabric mockups.
- Exploring virtual garment previews using recommended patterns and prints.
- Integrating the system with industrial cameras or production-line inspection equipment.
- Conducting further usability and performance testing with garment-sector users.

Conclusion

FabricIQ AI proposes an integrated AI-powered platform to assist fabric inspection, visual quality analysis, fabric appearance identification, design-reference recommendation, care-label interpretation, and automated reporting. By combining computer vision, OCR, image similarity, and recommendation techniques, the system aims to reduce repetitive manual tasks and make inspection information easier to review and manage.

The project will prioritize a feasible prototype with a limited number of supported defect and fabric categories, evaluated under controlled conditions. AI outputs will be treated as estimates and reviewed by human users. This approach will allow the team to demonstrate the practical use of AI in a garment-related workflow while clearly recognizing the system's technical and data limitations.